

Latin Square Designs

PUBH 526

Design and Analysis of Randomized Trials in Public Health

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Example

- Courtesy of Bruce Craig, Purdue Statistics
- Experiment to investigate the effect of 4 diets on cows' milk production
- There are 4 cows in this study, and each cow has 4 lactation periods
- The 4 diets are randomized across cows and lactation periods in a Latin square design
 - Each cow gets all 4 treatments (diets)
 - In every lactation period, one cow is receiving each of the 4 treatments (diets)
- Latin Square Cow Example.sas
- After each lactation period, there is a washout period so the previous diet does not affect the results of the next diet
- Cow and lactation period are blocks
- Treatment and lactation period are fixed effects. Cow is random (animals sampled from and intended to represent a larger herd).

Latin Square Design

- Suppose you want to block on two nuisance factors
- For this design, you need equal numbers of blocks and treatments
- Completely additive model (no interaction):

$$y_{ijk} = \mu + \alpha_i + \tau_j + \beta_k + \epsilon_{ijk} \quad \left\{ \begin{array}{l} i = 1, 2, \dots, p \\ j = 1, 2, \dots, p \\ k = 1, 2, \dots, p \end{array} \right.$$

α_i - i th Block 1 effect (row)

τ_j - j th treatment effect

β_k - k th Block 2 effect (column)

$\epsilon_{ijk} \sim N(0, \sigma^2)$

Layout Examples

Time / Rounds

Participants

C	B	A
B	A	C
A	C	B

B	A	C
A	C	B
C	B	A

D	B	C	A
C	D	A	B
B	A	D	C
A	C	B	D

Latin Square Design

- The two blocking factors are nuisance factors
 - Don't care if they're confounded, not really interested in statistical testing about them
 - Want to separate block effects from the treatment effects, which are of interest
 - Not anticipating/hypothesizing treatment*block effects (this can be relaxed)
- Design commonly represented in $p \times p$ grid

Latin Square Design

- There are now two randomization restrictions
 - One trt per row (row = blocking factor 1)
 - One trt per column (column = blocking factor 2)
- Can randomly shuffle rows, columns, and treatments of a standard square to get other variations of layout
 - Also, can search online / books or use SAS proc plan

Time / Rounds

Participants	D	B	C	A
	C	D	A	B
	B	A	D	C
	A	C	B	D

Analysis of Variance Table

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F_0
Rows	SS_{Row}	$p - 1$	MS_{Row}	
Treatment	$SS_{\text{Treatment}}$	$p - 1$	$MS_{\text{Treatment}}$	F_0
Column	SS_{Column}	$p - 1$	MS_{Column}	
Error	SS_E	$(p - 2)(p - 1)$	MS_E	
Total	SS_T	$p^2 - 1$		

If $F_0 > F_{\alpha, p-1, (p-2)(p-1)}$ then reject H_0

Analysis - Recommendations

- Used proc mixed (avoid proc glm)
- Use Type III tests, especially if missing values
- Use lsmeans rather than means

Mixed Model

- Often at least one block is considered random
 - Example: 4 cows randomly chosen from large herd
 - Want the inference to extend to the herd
- Sometimes time is one of the blocking factors
 - Example: Each cow measured over 4 lactation periods
 - This is an example of a crossover design
 - Could observations closer together be more correlated?
- SAS proc mixed can incorporate both concepts into the statistical model

“Latin Square Cow Example.sas”

- Why would cow or period affect milk yield?
- Useful rules:
 - Use Type III tests
 - Use lsmeans statement over means statement
- Diagnostic plots are not as informative because sample size is small

Correlated Errors

- Our observations are no longer completely independent
- Errors from same experimental unit (cow in this example) may be correlated
- Errors closer in time may be more similar
- Can incorporate various correlation structures into our models
- p observations per experimental unit results in a $p \times p$ covariance matrix
- Main diagonal contains the variances
- Off-diagonal elements represent covariances

Some Covariance Matrix Options

- Uncorrelated errors (for $p = 4$ obs/cow)

$$\begin{bmatrix} \sigma^2 & 0 & 0 & 0 \\ 0 & \sigma^2 & 0 & 0 \\ 0 & 0 & \sigma^2 & 0 \\ 0 & 0 & 0 & \sigma^2 \end{bmatrix} = \sigma^2 \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- 1st order autoregressive (for $p = 4$ obs/cow)

$$\sigma^2 \begin{bmatrix} 1 & \rho & \rho^2 & \rho^3 \\ \rho & 1 & \rho & \rho^2 \\ \rho^2 & \rho & 1 & \rho \\ \rho^3 & \rho^2 & \rho & 1 \end{bmatrix}$$

Some Covariance Matrix Options

- Compound symmetry (for $p = 4$ obs/cow)

Also called
“exchangeable”

$$\begin{bmatrix} \sigma^2 + \sigma_1 & \sigma_1 & \sigma_1 & \sigma_1 \\ \sigma_1 & \sigma^2 + \sigma_1 & \sigma_1 & \sigma_1 \\ \sigma_1 & \sigma_1 & \sigma^2 + \sigma_1 & \sigma_1 \\ \sigma_1 & \sigma_1 & \sigma_1 & \sigma^2 + \sigma_1 \end{bmatrix}$$

- Toeplitz (for $p = 4$ obs/cow)

$$\begin{bmatrix} \sigma^2 & \sigma_1 & \sigma_2 & \sigma_3 \\ \sigma_1 & \sigma^2 & \sigma_1 & \sigma_2 \\ \sigma_2 & \sigma_1 & \sigma^2 & \sigma_1 \\ \sigma_3 & \sigma_2 & \sigma_1 & \sigma^2 \end{bmatrix}$$

Some Covariance Matrix Options

- Banded Toeplitz $d = 2$ (for $p = 4$ obs/cow)

$$\begin{bmatrix} \sigma^2 & \sigma_1 & 0 & 0 \\ \sigma_1 & \sigma^2 & \sigma_1 & 0 \\ 0 & \sigma_1 & \sigma^2 & \sigma_1 \\ 0 & 0 & \sigma_1 & \sigma^2 \end{bmatrix}$$

- Unstructured (for $p = 4$ obs/cow)

$$\begin{bmatrix} \sigma_1^2 & \sigma_{12} & \sigma_{13} & \sigma_{14} \\ \sigma_{12} & \sigma_2^2 & \sigma_{23} & \sigma_{24} \\ \sigma_{13} & \sigma_{23} & \sigma_3^2 & \sigma_{34} \\ \sigma_{14} & \sigma_{24} & \sigma_{34} & \sigma_4^2 \end{bmatrix}$$

Back to “Latin Square Cow
Example.sas”

Extensions of the Latin Square

- Latin squares can have limited power / error degrees of freedom
- Extend by replicating the square (doing multiple squares, if you have the resources)
- Other variations exist

Latin Square Cow Example - With Confounding.sas